

Otmane El Bourki

AI Engineer · Production LLM Systems · Inference Optimization · Multi-Agent Orchestration

Carrières-sous-Poissy, Île-de-France · Currently in Rabat, Morocco · French residence permit
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AI Engineer who ships across the full LLM stack — retrieval and agentic orchestration, inference optimization, and production deployment — with a track record of measurable impact (**3× inference throughput at 47% lower cost**). Currently building group-wide AI accelerators at Alten's A3 (AI at Alten) program, following production delivery across finance and hospitality sectors.

Experience

AI Engineer — A3 (AI at Alten) June 2026 – Present
Alten — International Engineering & Technology Consulting Group Rabat, MA

- Member of **AI Management System & Accelerators**, a core team within **A3 (AI at Alten)**, the group-wide AI program run by Alten's Chief AI Office; building reusable AI components and solutions for deployment across Alten client projects.
- Shipped a production **reference-project AI search platform** now used by sales teams and Directeurs Techniques across the Alten group, reaching **81 active users since launch** with **~600 ms average response time**; deployed on Azure App Service with CI/CD via Azure DevOps.
- Built an end-to-end **LLM ingestion pipeline** using Mistral OCR and structured-output generation to transform unstructured PPTX project decks into schema-conformant JSON, populating a searchable project-reference database.
- Designed the **retrieval and generation layer** using Mistral embeddings for semantic search and LLM-generated natural-language summaries, with OpenAI as a fallback for generation reliability.

AI Engineer Feb. 2026 – May 2026
Independent — Self-directed R&D & Open Source Paris, FR

- Open-sourced **codibox** (PyPI), a Docker-sandboxed Python execution engine enabling safe AI-generated code execution for LangGraph and CrewAI.
- Designed a **multi-agent real estate platform** with LangChain deepagents for valuation, ROI, and market research, cut research time from **~3 hours to 15 minutes**.
- Built a **GraphRAG platform** on Neo4j with hybrid vector/full-text retrieval for LLM-powered due diligence and compliance Q&A.

AI Engineer (CDD) Feb. 2025 – Feb. 2026
Impactera — Financial Consulting Paris, FR (Remote)

- Architected a supervisor-based **multi-agent platform (LangGraph, 6 subagents)** spanning document **RAG**, live API/SQL access, web research, and report generation, cut report generation from **half a day to 30 minutes**.
- Built a self-correcting **RAG pipeline (Qdrant)** with relevance-graded retrieval and hallucination-checked answers that auto-fallback on failed validation, shaped by CEO and finance-consultant requirements.
- Engineered **multi-LLM routing** (OpenAI, Anthropic, Gemini, Bedrock) with fallback and async checkpointing, cutting rate-limit failures to **near zero**.
- Built a nested **API-orchestration agent** that constructs tool-calling agents at runtime from **OpenAPI specs** (Tagetik and client systems).

AI Engineer (CDD) June 2023 – June 2024
Tetricks — AI Solutions for Hospitality Remote

- Led Vision-LLM integration for automated hotel room description generation, working directly with the CEO and Dev team; PoC **secured Bpifrance innovation funding**.
- Engineered a **vLLM serving stack** with AWQ/GPTQ quantization and dynamic batching, achieving **3× throughput** and **47% cost reduction** vs. baseline serving.
- Fine-tuned **Llama 2** on hospitality domain data for guest intent detection across **12+ intent categories**, deployed into production guest-messaging workflows.
- Built a multimodal **RAG chatbot** on GCP (Vertex AI + Cloud Run) with a Pydantic-enforced ingestion pipeline and hybrid vector/lexical retrieval with reranking.

ML Research Assistant (Academic, Voluntary) Jan. 2022 – June 2023
Ibn Tofail University — Research Lab Kenitra, MA

- Developed a **medical RAG chatbot** (Meditron 7B, Qdrant, PubMedBERT) for evidence-grounded clinical Q&A, reducing hallucination risk via source-cited retrieval.
- Built **North**, a dual-stack North African dialect classifier (**scikit-learn + fine-tuned BERT**) across Arabic and Arabizi Latin script, achieving **94.5% test accuracy**.
- Benchmarked SOTA CV architectures on FER2013, achieving **90% avg. precision** and **20% generalization improvement** via targeted augmentation, regularization.

Leadership & Community

Founder & President Oct. 2019 – Jan. 2022
1337 AI — Educational Organization Benguerir, MA

- Founded and scaled a **300-member AI/ML community**, organizing 2 bootcamps and 6 workshops on Python, ML, and GenAI.
- Co-mentor at **ThinkAI 2023** and **HackAI 2024 & 2025**, Morocco's first generative-AI hackathons, alongside mentors from Hugging Face and Mistral AI.

Technical Skills

LLM Orchestration & Agentic Systems

LangGraph LangChain CrewAI Multi-Agent Systems
ReAct / Plan-Execute Tool Calling MCP
Structured Outputs

Retrieval & Knowledge Systems

RAG GraphRAG Hybrid Search Reranking
Semantic Chunking Qdrant Neo4j OCR
Multimodal Document Understanding

LLM Providers & Model Integration

OpenAI Anthropic Gemini Mistral Hugging Face
Multi-LLM Routing

Evaluation, Observability & Responsible AI

RAGAS LangSmith LLM-as-Judge Hallucination Detection
EU AI Act & Risk Assessment Bias & Safety Evaluation

Inference & Production Engineering

vLLM Quantization (AWQ/GPTQ) Fine-Tuning (LoRA/QLoRA)
Rate Limiting & Fallback Strategies Cost & Latency Optimization

AI Architecture & Deployment

End-to-End AI System Design Vertex AI Amazon Bedrock
Azure AI Docker Kubernetes Terraform
CI/CD MLflow AWS GCP Azure

Programming & APIs

Python FastAPI Pydantic SQL

Education

IT Architecture Expert: Data Architecture — RNCP Level 7 (Equiv. Master's)

École 42 Paris | Level 21+ (Top Tier of Global Curriculum) · 3rd globally in pedagogical innovation (WURI 2025)
Foundation studies at École 1337, UM6P

Master's in Artificial Intelligence

Ibn Tofail University

BSc (Licence) in Computer Science

Ibn Tofail University

Projects

Systems Depth for AI Engineering

École 42 Paris, Oct. 2024–Feb. 2025 · GPU/ML foundations underlying production inference work
• Built a CUDA-accelerated C/C++ linear algebra engine (matrix operations, inverses) — the same class of GPU-level primitives underlying LLM inference kernels.
• Developed a real-time ML-based BCI pipeline (EEG signal classification via CSP/PCA feature extraction), achieving **>90% accuracy** with **<2 s** end-to-end inference latency.

Certifications

GCP AI Infrastructure Google Cloud 2026
GPU Programming Johns Hopkins 2026
GenAI Winter School École Polytechnique 2024
Red Teaming LLM Apps DeepLearning.AI 2025
LangGraph LangChain 2025
Advanced RAG DeepLearning.AI 2024
Deep Learning Specialization DL.AI 2020

Open Source

codibox github.com/oelbourki/codibox · PyPI
Languages

Arabic (native) French (B2) English (C2)